BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: PR XI (01) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A — Model Paper

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** Biology Paper–XI (Model Paper)

Note: CAVEAT: the objective sheet in the source is headed "Model Paper · Biology Paper – XI" (no year) — it is NOT the board's genuine 2021 objective sheet; kept as supplied.

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. All are organelles except.....

(A) Endoplasmic reticulum
(B) Lysosome
(C) Cell membrane
(D) Golgi complex

Q2. Single membrane bounded organelle is .....

(A) Mitochondria
(B) Nucleus
(C) Chloroplast
(D) Lysosome

Q3. The monomer of which polysaccharide possesses nitrogen group?

(A) Glycogen
(B) Chitin
(C) Starch
(D) Cellulose

Q4. Which type of Lipid Play role in maintainance of male and female characteristics in Human?

(A) Triglyceride
(B) Steroids
(C) Waxes
(D) Phospholipids

Q5. Enzymes action in living cell changes with change in .....

(A) H^+ ion concentration
(B) Na^+ ion concentration
(C) OH^- ion concentration
(D) Both A & C are correct

Q6. Plants are the lungs of the nature because it provide.....

(A) Food
(B) Oxygen
(C) Carbon dioxide
(D) Humidity

Q7. In the absence of oxygen which one is the product of respiration in animal?

(A) Acetic Acid
(B) Ethyl alcohol
(C) Lactic Acid
(D) Oleic Acid

Q8. All are correct about viroids except one.

(A) Posses double stranded (RNA)
(B) Posses single stranded RNA
(C) Lack protein coat
(D) Smaller than virus

Q9. One is the alternate of mitochondria in bacterial cell.

(A) Cell membrane
(B) Ribosome
(C) DNA
(D) Cell wall

Q10. A nitrogen containing polysaccharide is present in .....

(A) Plant cell wall
(B) Bacterial cell wall
(C) Fungal cell wall
(D) Viral envelop

Q11. In sunflower plant all the sessile flowers are crowded together on short axis called.....

(A) Umbell
(B) Panicle
(C) Corymb
(D) Capitulum

Q12. Among plants which group is called invader of land.

(A) Gymnosperm
(B) Angiosperm
(C) Pteridophyte
(D) Bryophyte

Q13. All are metamerically segmented except .....

(A) Earthworm
(B) Neries
(C) Leech
(D) None of these

Q14. Which one of the given belong to phylum coelentrata?

(A) Lung fish
(B) Cuttle fish
(C) Jelly fish
(D) Star fish

Q15. In which photocycle long day plants will give flower?

(A) 15 hour light / 1 hour dark
(B) 8 hour light / 12 hour dark
(C) 10 hour light / 14 hour dark
(D) 9 hour light / 15 hour dark

Q16. The function of hairs on stem and leaves of desert plant is .....

(A) Protection
(B) To Retard transpiration
(C) To Repel insect
(D) Storage of food

Q17. Amylase hydrolyses starch into

(A) Lactose
(B) Maltose
(C) Fatty acid
(D) Galactose

Q18. A cardiovascular disorder causes due to lack of blood supply to heart muscle is called.....

(A) Arteriosclerosis
(B) Cyanosis
(C) Angina pectoris
(D) Cardiac arrest

Section-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

QII. Answer any TEN parts. Each part carries FOUR marks.

1. Write a short note on cytoskeleton.
2. Describe the properties of Monosaccharides.
3. How temperature affects the rate of enzyme actions?
4. Briefly mention the mechanism of Calvin cycle.
5. Describe some control measures against the transmission of HIV.
6. Discuss the autotrophic mode of nutrition in bacteria.
7. Give an account of the class Basidiomycota.
8. Write differences between Gymnospermae and Angiospermae.
9. Mention some distinguishing characters of phylum Arthropoda.
10. Explain movements of water in xylem through TACT mechanism.
11. Differentiate between the structure and function of small and large intestine.
12. Discuss the structure of human heart.

13. Write a short note on acquired immunity.

Section-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 27

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [4 ½] Describe the types of proteins on the basis of its shapes.

(b) [4 ½] Explain the lytic cycle of Bacteriophage.

QIV.

(a) [5] Classify bacteria on the basis of shape.

(b) [4] Write two similarities and two differences between mitochondria and chloroplast.

QV.

(a) [5] Discuss the mechanical and chemical digestion in the oral cavity.

(b) [4] Write any four importances of Fungi.

QVI. Write notes on any two of the following.

(a) [4 ½] Functions of lymphatic system

(b) [4 ½] Role of macrophages

(c) [4 ½] Macronutrients